LARS KROEGER

Zürich, Switzerland

Available from May 2027 – flexible start date and duration.

EDUCATION

ETH Zurich – M.Sc. Electrical Engineering and Information Technology — Expected early 2028
Electronics and Photonics. Current weighted average: **5.70/6.00**

ETH Zurich – B.Sc. Electrical Engineering and Information Technology — Feb. 2025
Final grade: **5.66/6.00**

University of Texas at Austin – Exchange semester. GPA: **3.83/4.00** — Spring 2024

PROJECTS

NoC Architecture and Link-Width Scaling for a Many-Core RISC-V Accelerator — Semester Project
SystemVerilog, Synopsys Fusion Compiler, TSMC 7 nm — *2026*

- Developed a parameterized AXI width adapter to decouple DMA and NoC widths and a source-ID-aware round-robin read interleaver to reduce burst serialization, achieving 2× baseline performance on memory-bound benchmarks with 1024-bit NoC links and 16 active clusters.
- Evaluated area, timing, and routing limits in TSMC 7 nm using Synopsys Fusion Compiler.
- Optimized FlashAttention-2 decode using bank-aware placement and SIMD for a 17.6× single-cluster compute speedup. Used the kernel to demonstrate 1.8× speedup from widening DMA/NoC datapaths from 512 to 1024 bits.

RISC-V SoC Design and ASIC Implementation — University Project
SystemVerilog, Yosys, OpenROAD

- Co-developed a general-purpose DMA engine and four-mode SPI controller for the Croc RISC-V SoC, achieving 1.75× graphics-demo speedup through overlapping computation and DMA transfers.
- Wrote the complete RTL-to-GDSII flow using Yosys and OpenROAD. Our team was one of five selected for fabrication, with successful first-silicon bring-up.

Federated Continual Learning on RISC-V SoC for Nano-Drone Swarms — Bachelor's Thesis

- Implemented multicore on-device DNN training and inference on GAP9 with UWB model synchronization across nano-drones, achieving 178 ms and 4.3 mJ per local training epoch in low-power mode.
- First author of work presented at RISC-V Summit Europe 2025. [Paper]

Standard Cell Library Design and Characterization — University Project
EDA Tools: Cadence, Synopsys

- Contributed standard cells to the open-source EZ130 8T library in IHP 130 nm. Co-author of the associated paper. [Paper]

WORK EXPERIENCE

Enclustra GmbH – FPGA/SoC Solutions — Zurich, Switzerland
Hardware Engineering Intern – Innovation Lab — *Oct. 2025 – Mar. 2026*

- Developed and validated FPGA/SoC hardware, including AMD Versal platforms.
- Built Python and Altium tools to automate schematic/netlist checks and component-database workflows.

TEACHING AND DISTINCTIONS

Teaching Assistant, ETH Zurich – Calculus and Digital Circuits — 2022–2024
Received the **Outstanding Teaching Assistant Award**, 2024. [Award]
Swiss Study Foundation – Member since 2022. Selective academic excellence and leadership programme. [Website]

TECHNICAL SKILLS

RTL / SoC	SystemVerilog, Verilog, AXI, DMA, NoC, RISC-V
EDA Tools	Synopsys Fusion Compiler / Design Compiler, Cadence Innovus, Yosys, OpenROAD
Programming	Python, C, C++, Tcl, Bash, MATLAB
FPGA / ML	Vivado, Vitis HLS, PyTorch, embedded ML in C